

Gauge Transformation of a Plane Electromagnetic Wave

D. Tong, Electromagnetism, Problem Sheet 3, Question 5.

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Electrodynamics

Consider a plane polarized electromagnetic wave described by the vector and scalar potentials,

$$\mathbf{A}(\mathbf{r}, t) = \mathbf{A}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad \text{and} \quad \phi(\mathbf{r}, t) = \phi_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}, \quad (1)$$

with constant $\mathbf{A}_0$ and ϕ_0 . Use Maxwell's equations to find a relationship between $\mathbf{A}_0$ and ϕ_0 .

Find a gauge transformation such that the new vector potential is “transversely polarised”, i.e. $\mathbf{A}_0 \cdot \mathbf{k} = 0$ (Coulomb gauge, $\nabla \cdot \mathbf{A} = 0$). What is the scalar potential ϕ in this gauge?

Solution The Maxwell equations give

$$\nabla^2 \phi + \frac{\partial}{\partial t} \nabla \cdot \mathbf{A} = 0, \quad (2)$$

and

$$\nabla \times (\nabla \times \mathbf{A}) = -\frac{1}{c^2} \left[\nabla \left(\frac{\partial \phi}{\partial t} \right) + \frac{\partial^2 \mathbf{A}}{\partial t^2} \right] = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}. \quad (3)$$

Then, we find

$$\mathbf{k}^2 \phi_0 - \omega \mathbf{k} \cdot \mathbf{A}_0 = 0. \quad (4)$$

If we set the Coulomb gauge, we find

$$\phi_0 = 0. \quad (5)$$